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1 Task 1: Resonance Frequency Analysis for Wire Resistor Circuits

1.1 Theory

The inductance of a $R = 12\ \Omega$ wire resistor can be determined by measuring the series or parallel resonance frequency of the given circuits. We determined the resonant frequencies using the following formulas:

Circuit 1: Series Resonance

$$f_{1,\text{Res}} = \frac{1}{2\pi\sqrt{L_S \cdot C}}$$

Circuit 2: Parallel Resonance

$$f_{2,\text{Res}} = \frac{1}{2\pi} \sqrt{\frac{1}{L_S \cdot C} - \frac{R^2}{L_S^2}}$$

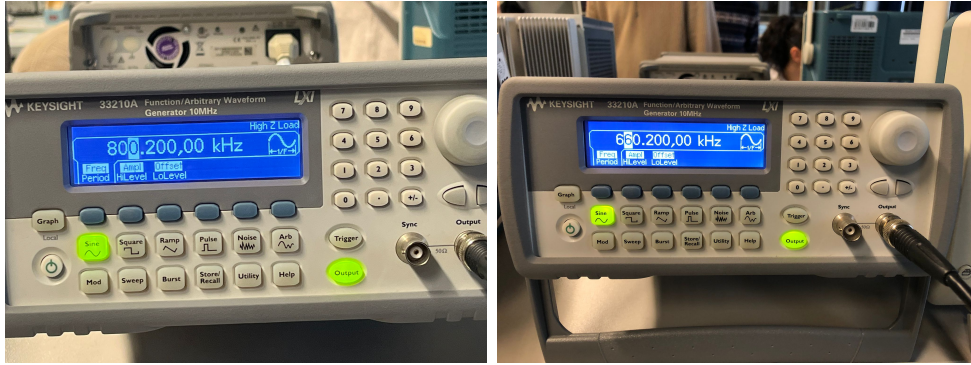


Figure 1: Circuit 1 and 2 Measured Resonant Frequencies

1.2 Calculations

Given parameters:

- Capacitance: $C = 10\ \text{nF} = 10 \times 10^{-9}\ \text{F}$
- Resistance: $R = 12\ \Omega$
- Resonant frequency for Circuit 1, calculated in the oscilloscope: $f_{1,\text{Res}} = 800.200\ \text{Hz}$
- Resonant frequency for Circuit 2, calculated in the oscilloscope: $f_{2,\text{Res}} = 660.200\ \text{Hz}$

Using the formula for Circuit 1, the inductance L_S is calculated as:

$$L_S = \frac{1}{(2\pi f_{1,\text{Res}})^2 \cdot C} = \frac{1}{(2\pi \cdot 800 \times 10^3)^2 \cdot 10 \times 10^{-9}} \approx 3.96\ \mu\text{H}$$

For Circuit 2, substituting L_S and C into the formula:

$$f_{2,\text{Res}} = \frac{1}{2\pi} \sqrt{\frac{1}{L_S \cdot C} - \frac{R^2}{L_S^2}}$$
$$f_{2,\text{Res}} = \frac{1}{2\pi} \sqrt{\frac{1}{3.96 \times 10^{-6} \cdot 10 \times 10^{-9}} - \frac{12^2}{(3.96 \times 10^{-6})^2}} \approx 638.1 \text{ kHz}$$

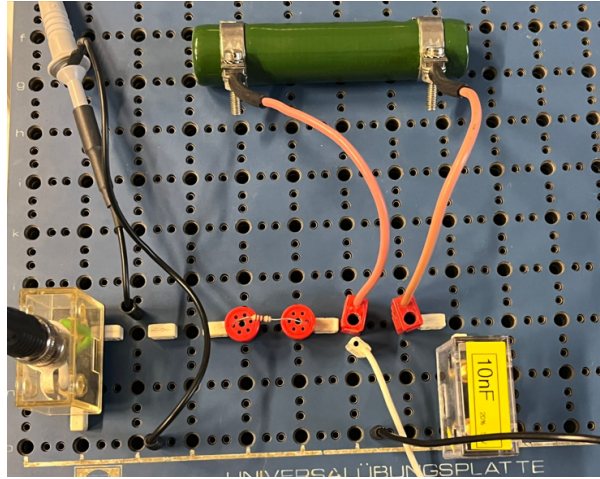


Figure 2: Our lab setup for Circuit 1

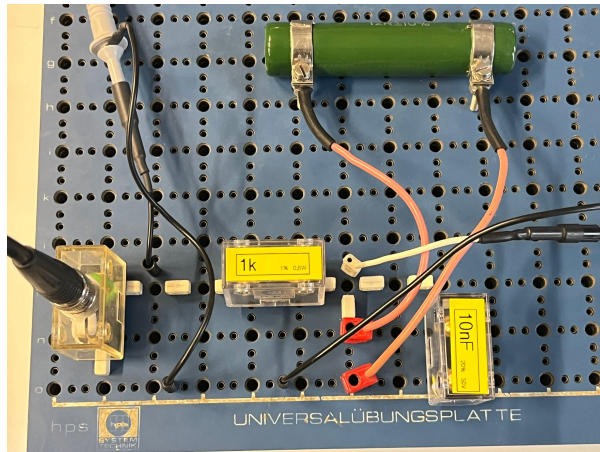


Figure 3: Our lab setup for Circuit 2

1.3 HP 4294A Impedance Analyzer

We used the Impedance Analyzer in the lab in order to calculate the exact value of the Capacitor.

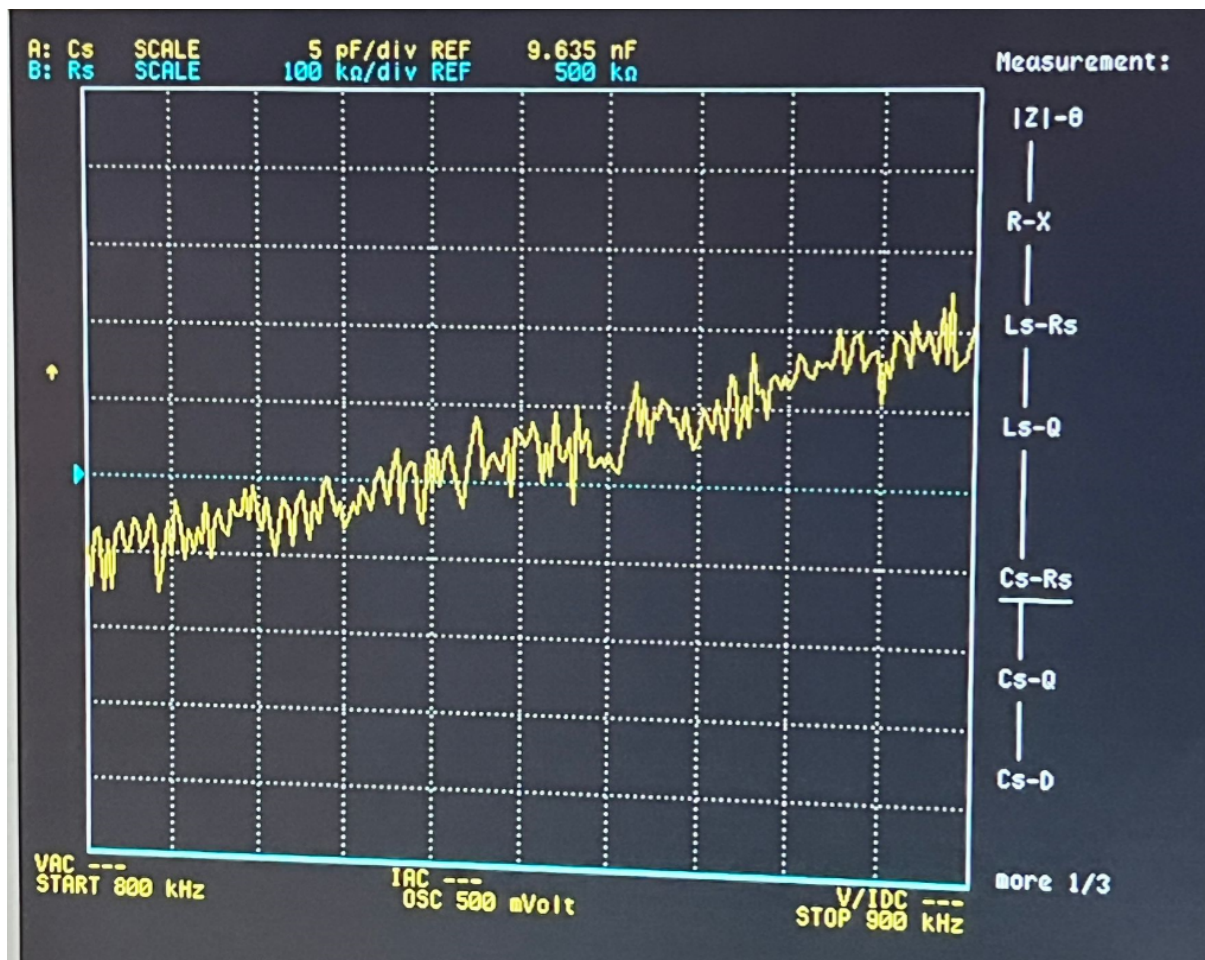


Figure 4: Exact Value Calculated via Impedance Analyzer

2 Task 2: Loss Resistance of an Inductance

2.1 Objective

In this exercise, the parasitic resistance of an $L = 22\text{ mH}$ inductance is investigated. The series loss resistance is determined by setting the circuit in series resonance condition by using a low-loss capacitor (Capacitance Decade) at a given measuring frequency.

At resonance, the reactances of L and C compensate each other, so R_s can be calculated from the ratio of U_1 and U_2 .

2.2 Circuit Setup

The circuit setup for this task is shown in Figure 5.

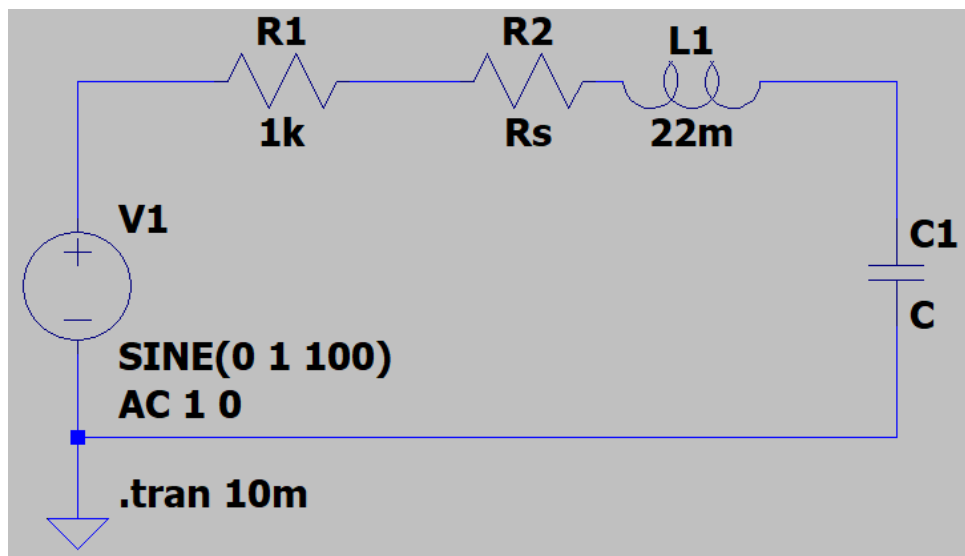


Figure 5: LTspice model in order to calculate loss(series) resistance

2.3 Preparation

- Calculate the theoretical values of C in resonance condition for $f = 1, 2, 5, 10, 20, 50, 100\text{ kHz}$ using the resonance formula:

$$C = \frac{1}{(2\pi f)^2 \cdot L}.$$

- Use these values as the initial setting of C in the lab session.

2.4 Lab Tasks

- Measure the DC resistance of the inductor.
- Measure U_1 and U_2 for the following frequencies: $f = 1, 2, 5, 10, 20, 50, 100\text{ kHz}$.

2.5 Calculation of R_s

The series loss resistance R_s can be calculated using the measured voltages U_1 and U_2 as follows:

$$R_s = R \cdot \frac{U_2}{U_1 - U_2}.$$

Here, $R = 1 \text{ k}\Omega$ is the known series resistance in the circuit.

2.6 Capacitance vs Frequency Analysis

The calculated capacitance values for the resonance condition at different frequencies are tabulated below:

Frequency f (kHz)	Capacitance C (nF)
1	1151.38
2	287.84
5	46.06
10	11.51
20	2.88
50	0.46
100	0.12

Table 1: Theoretical capacitance values at different frequencies

The relationship between capacitance and frequency is plotted in a log-log scale as shown in Figure [2.6](#).

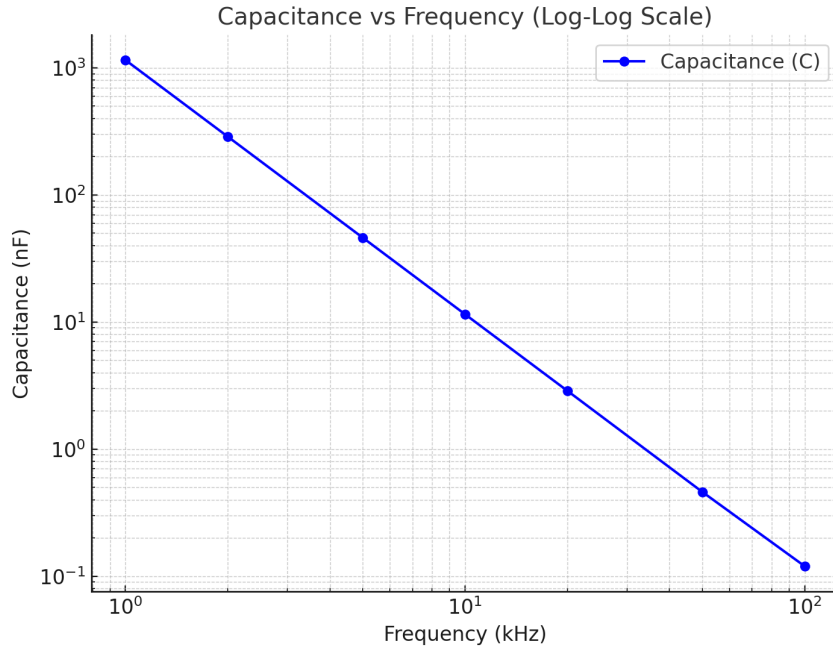


Figure 6: Capacitance vs Frequency (Log-Log Scale)

2.7 DC and AC Resistance Measurement of the Inductor

The resistance of the inductor was measured at different frequencies using a multimeter and impedance analyzer. These measurements help evaluate the inductor's parasitic effects.

Measurement at 100 Hz

The measurement setup and results for 100 Hz are shown in Figure 2.7.

Measurement at 500 Hz

The measurement setup and results for 500 Hz are shown in Figure 2.7.

Measurement at 1 kHz

The measurement setup and results for 1 kHz are shown in Figure 2.7.

2.8 U_1 and U_2 Measurements

The values of U_1 and U_2 were measured at different frequencies. These measurements are calculated in Table 2.8.

The variation of U_1 and U_2 with frequency is plotted in Figure 2.8.

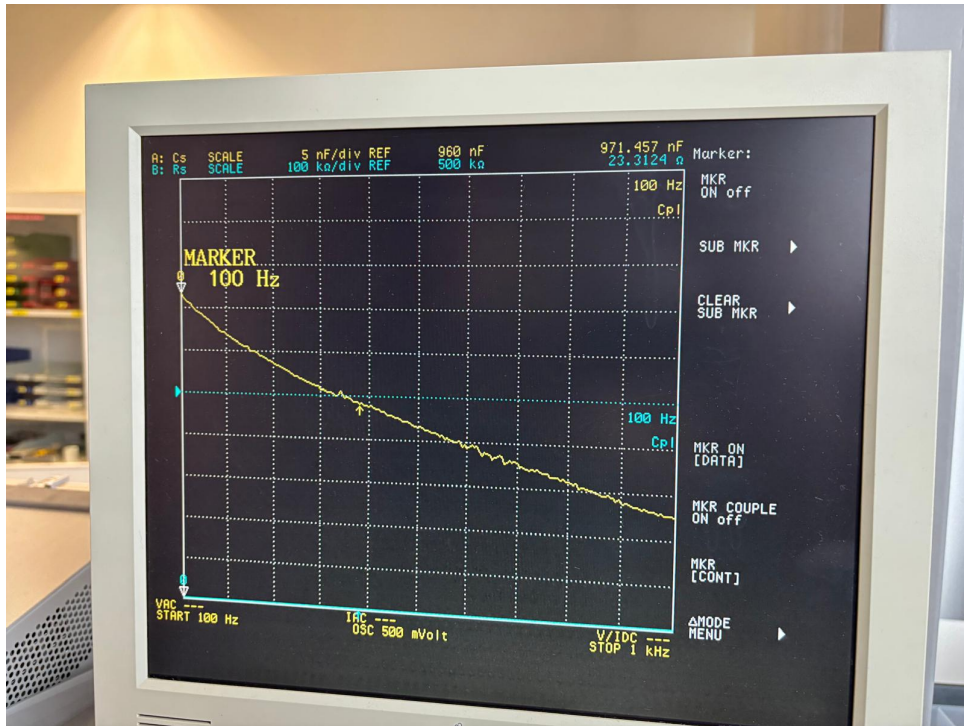


Figure 7: Resistance measurement with HP 4294A Impedance Analyzer of the inductor at 100 Hz

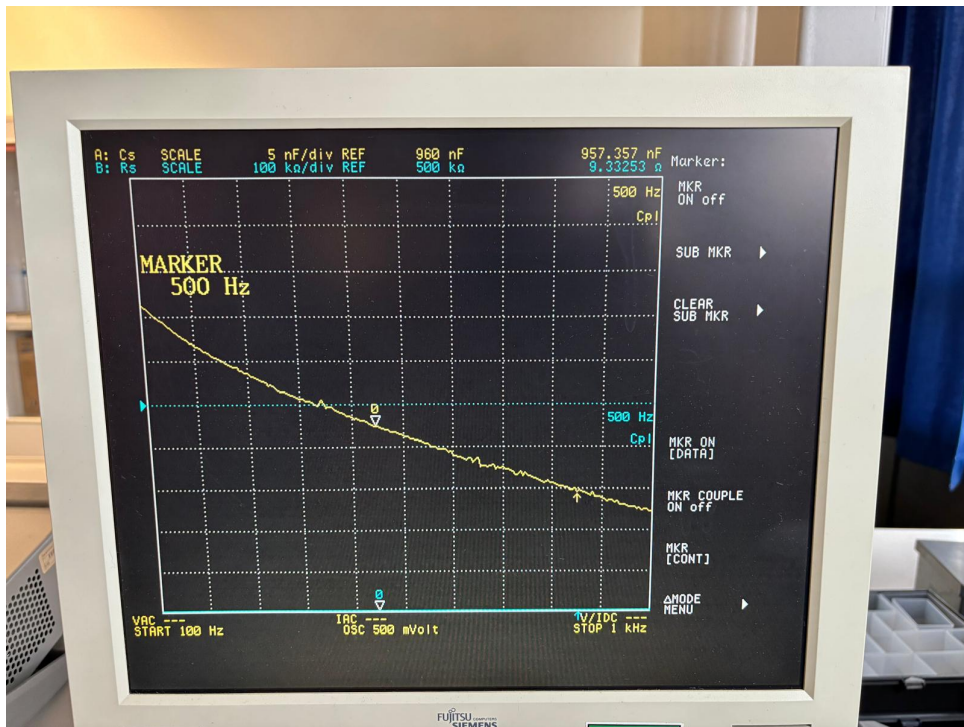


Figure 8: Resistance measurement with HP 4294A Impedance Analyzer of the inductor at 500 Hz

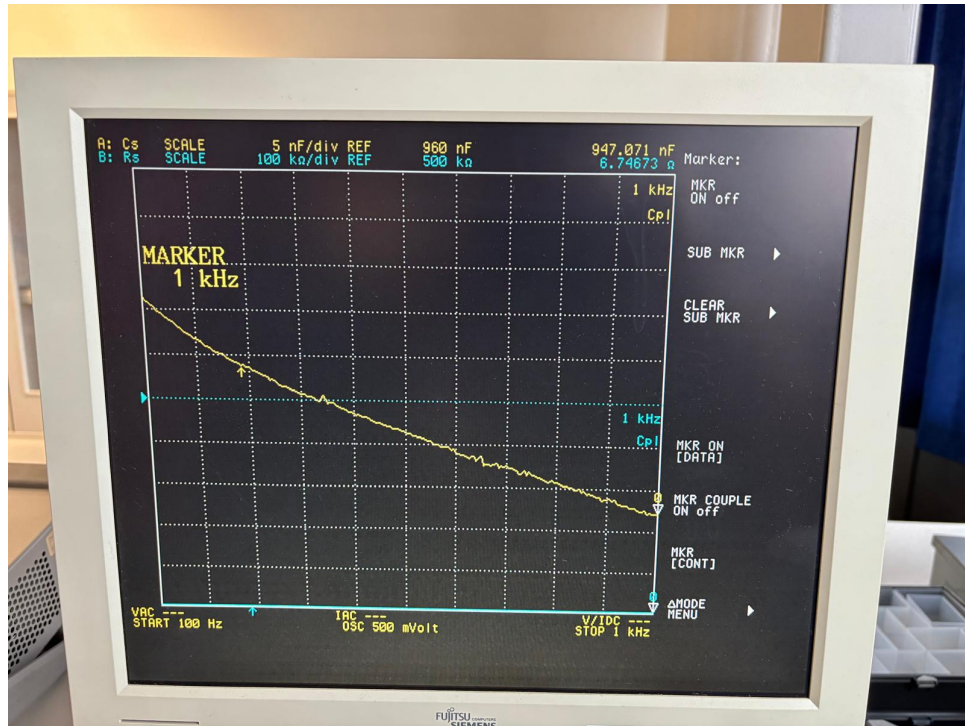


Figure 9: Resistance measurement of the inductor at 1 kHz

Frequency f (kHz)	U_1 (mV)	U_2 (mV)
1	920	36.9
2	914	45.3
5	907	38.6
10	908	65.0
20	909	168
50	903	700
100	952	985

Table 2: Measured values of U_1 and U_2 at different frequencies

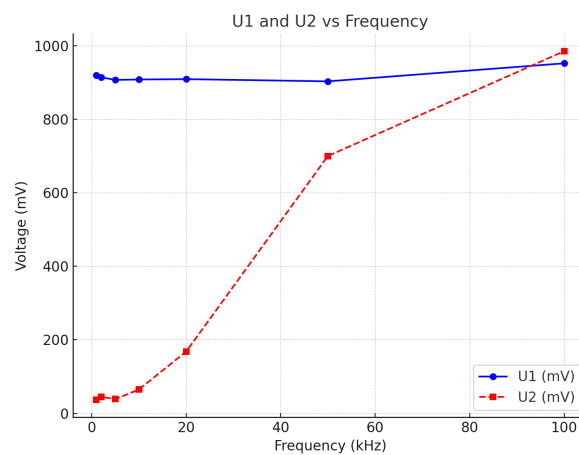


Figure 10: Variation of U_1 and U_2 with frequency

3 Series Loss Resistance of a Capacitor

3.1 Theory

The series loss resistance (ESR) of a $C = 1\ \mu\text{F}$ electrolytic capacitor is measured using a bridge circuit, as illustrated in Figure 11. To achieve a balanced bridge network ($U_{\text{Br}} = 0$), the following relations apply:

$$\frac{C_n}{C_x} = \frac{R_1}{R_2}, \quad \frac{R_n}{R_x} = \frac{R_2}{R_1}. \quad (1)$$

In this experiment, the resistor R_n and the capacitance C_n are adjusted alternately until no change in the bridge voltage U_{Br} is observed.

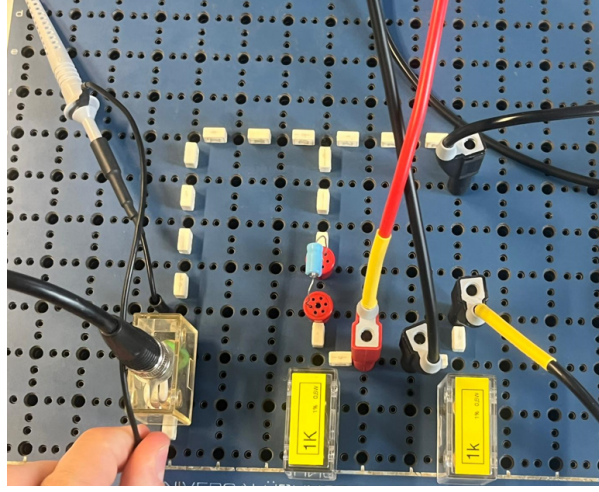


Figure 11: Bridge network for determining the capacitance and series loss resistance of an electrolytic capacitor

3.2 Experimental Procedure

- U_{Br} was measured using an oscilloscope triggered by U_{in} (Channel 1).
- The capacitance C_x and the series loss resistance R_x were measured at frequencies of 100 Hz, 500 Hz, and 1 kHz using the HP 4294A Impedance Analyzer.

3.3 Results

The measured values of capacitance C_x and series loss resistance R_x at different frequencies are presented in Table 3.3.

Frequency (Hz)	Capacitance C_x (nF)	Series Resistance R_x (Ω)
100	9640	2390
500	9540	990
1,000	9430	690

Table 3: Measured capacitance and series loss resistance at various frequencies

3.4 Oscilloscope Signals

The following graphs represent the oscilloscope signals captured at different frequencies:

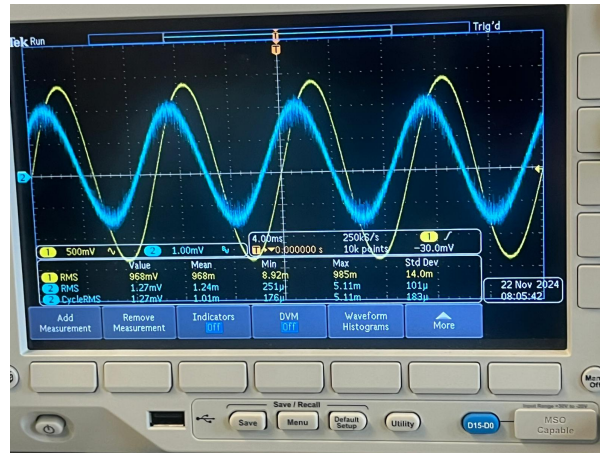


Figure 12: Oscilloscope signal at 100 Hz



Figure 13: Oscilloscope signal at 500 Hz

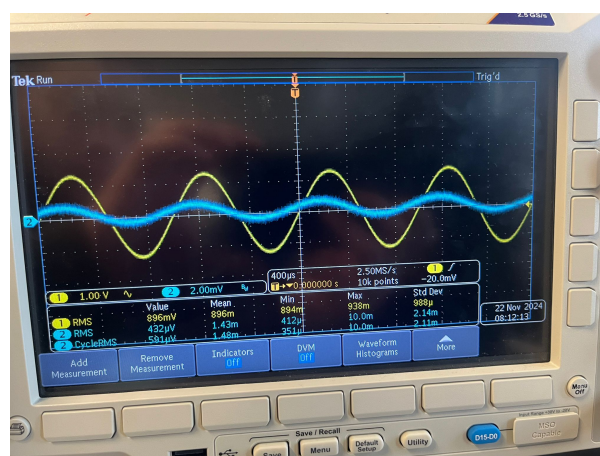


Figure 14: Oscilloscope signal at 1 kHz

3.5 Evaluation

The results for C_x and R_x obtained from the bridge network method were compared to those directly measured by the impedance analyzer. The comparison validates the experimental setup and highlights any deviations at different frequencies.

4 Summary of Lab Tasks

Task 1: Series Inductance of a Wire Resistor

The goal was to determine the series inductance L_S of a 12Ω wire resistor by measuring the resonance frequency in two circuits:

1. Circuit 1: A series LC circuit where $f_{1,\text{Res}} = \frac{1}{2\pi\sqrt{L_S C}}$.
2. Circuit 2: A parallel LC circuit with resistance, where $f_{2,\text{Res}} = \frac{1}{2\pi} \sqrt{\frac{1 - \frac{R^2}{L_S C}}{L_S C}}$.

Theoretical and experimental resonance frequencies were determined, and the results were compared to calculate L_S .

Task 2: Loss Resistance of an Inductance

The series resistance (R_S) of a 22 mH inductance was investigated. The resonance condition was achieved by setting an LC circuit at various frequencies (1, 2, 5, 10, 20, 50, 100 kHz). At resonance, R_S was calculated using voltage measurements U_1 and U_2 and plotted as $R_S = f(f)$ on a double logarithmic scale.

Task 3: Series Loss Resistance of a Capacitor

The Equivalent Series Resistance (ESR) of a $C = 1\mu\text{F}$ electrolytic capacitor was determined using a balanced bridge circuit. The capacitance C_X and the ESR (R_X) were measured at $f = 100\text{ Hz}$, 500 Hz , and 1 kHz by adjusting the resistors and capacitors until the bridge voltage $U_{\text{BR}} = 0$. The results were validated with an impedance analyzer.